Muhammad Hamza

AI Systems Engineer | Generative AI | Computer Vision | Agentic AI

Email: Hamza.ai.official@gmail.com ☎ WhatsApp: +92 302 4842593 ● Lahore, Pakistan

LinkedIn: linkedin.com/in/hamzzaz | **GitHub:** github.com/Hamza-code-hub | **Portfolio:** hamza-code-hub.github.io

Kaggle: kaggle.com/muhammadhamzamaher | **Fiverr:** fiverr.com/maher_hamza4616 | **Google Scholar:** scholar.google.com/citations

AI Engineer (SE) at CodingCops with approximately 3 years of experience in Agentic AI, LLM/RAG applications, Computer Vision, Machine Learning, and defensive cybersecurity. Previously led a four-person team at CyberZeus, owning client-facing AI delivery from requirements and architecture through evaluation, integration, and deployment. Co-author of four peer-reviewed journal papers spanning medical imaging and adversarially robust UAV security.

CORE TECHNICAL SKILLS

Agentic AI & LLM Engineering: LLMs, Deep Agents, Agent Harnesses, LangChain, LangGraph, Multi-Agent Workflows, Tool Calling, State & Memory Management, MCP, Prompt Engineering
ML, Deep Learning & Computer Vision: Machine Learning, Deep Learning, NLP, YOLO, OpenCV, ByteTrack, Object Detection & Tracking, CNNs, Transformers
RAG & Retrieval Systems: RAG Architecture, Embeddings, Document Chunking & Indexing, Semantic & Vector Search, FAISS, ChromaDB, LlamaIndex, Retrieval Evaluation
Programming, Backend & Data: Python, FastAPI, REST APIs, NumPy, Pandas, SQL, PostgreSQL, DynamoDB, Memgraph
Cloud, DevOps & Deployment: AWS, Docker, Kubernetes, GitHub Actions, CI/CD, Git
Cybersecurity & VAPT: Nmap, Network Reconnaissance, Port Scanning, OS/Service Fingerprinting, DNS/DNSSEC Analysis, CVE/KEV/EPSS Enrichment, OWASP API Top 10, Web/API Vulnerability Assessment, SNMP, SSL/TLS Auditing, JWT/Session Security, PCAP Analysis

EXPERIENCE

CodingCops

AI Engineer (SE) – CodingCops 10/2026 – Present
• Built and deployed production-grade AI applications using Deep Agents and LangGraph for multi-step planning, stateful orchestration, and tool-driven workflows.
• Developed agent harness components for tool interfaces, controlled execution, context management, guardrails, and repeatable agent evaluation.
• Worked with Memgraph to model and traverse connected data for graph-aware AI workflows and with DynamoDB for application data persistence.
• Integrated LLM-powered agents with Python APIs and backend services, connecting retrieval, graph context, and business logic in deployable systems.
• Collaborated on reliability, debugging, testing, and production integration of AI services and complex agent workflows.

CyberZeus

AI Engineer – Cyberzeus Software Systems 01/2026 – 10/2026
• Delivered client-facing AI/ML solutions end to end, from requirements and architecture through development, evaluation, deployment, and support; led technical discussions and demos.
• Led a four-person team (two junior developers and two interns), mentoring engineers, planning tasks, reviewing implementations, and coordinating integration.
• Architected LangGraph multi-agent and RAG cybersecurity workflows, integrating tool execution, vulnerability intelligence, evidence correlation, and remediation reporting.

AI Developer – Cyberzeus Games 08/2025 – 12/2025
• Designed, developed, and fine-tuned AI and machine-learning models for web-based and software products, supporting model integration, evaluation, testing, and iterative improvement.
• Applied Python-based AI development and intelligent-agent concepts to product features and engineering workflows in a game/software environment.

Minhaj Solutions, *AI Developer* 04/2024 – 08/2025
• Built an agentic RAG-based support assistant using LLM reasoning to understand and resolve technical product queries.
• Designed the knowledge pipeline from document chunking and embeddings through vector indexing, semantic retrieval, context selection, and response generation.
• Implemented retrieval-augmented, context-aware query handling to produce real-time answers grounded in product knowledge instead of relying on fixed rule-based flows.
• Contributed to computer-vision workflows covering image preprocessing, object detection/classification, model inference, evaluation, and backend integration for client-facing AI features.

Fiverr, *Freelance AI/ML Engineer & Python Developer* 07/2023 – Present
• Delivered 12+ international AI/ML projects in RAG, AI agents, automation, data analysis, and computer vision for clients.
• Built integrated AI workflows with Python, FastAPI, LangChain/LangGraph, OpenAI/Gemini, vector databases, PostgreSQL, REST APIs, and Docker.
• Maintained a 4.8/5 Fiverr rating across 13 reviews, reflecting delivery quality, communication, and repeat collaboration.

PROJECTS

AI Agents for Cybersecurity Operations & Vulnerability Intelligence

Problem solved: analysts must correlate fragmented network, web, API, CVE, and compliance evidence to prioritize remediation.

- Built a security operations platform with LangGraph/LiteLLM agents, RAG enrichment, FastAPI orchestration, Go/gRPC execution, and a Flutter desktop for scans, findings, vulnerabilities, reports, and AI-assisted analysis.
- Implemented host discovery, port/OS/service/DNS checks, CVE enrichment (KEV/SSVC/EPSS), OWASP API and web tests, SNMP, SSL/TLS, JWT/session security, PCAP forensics, and compliance assessment.
- Designed agentic tool selection and evidence correlation to merge live scan outputs with retrieved knowledge, deduplicate findings, prioritize risk, generate remediation reports, and stream/persist execution history.

Orbital RF Mission & Distributed Edge Operations Platform

Problem solved: satellite missions require dependable coordination of backend, desktop, edge agents, SDR hardware, fleet state, and visual evidence.

- Engineered the end-to-end architecture connecting a Flutter desktop client, Python APIs/services, queued commands, distributed Linux edge agents, telemetry/state, and controlled payload execution.
- Stabilized authenticated LimeSDR half-duplex RF delivery: identity/key integrity, RX timing, SDR contention, false-success states, terminal status, fleet lifecycle, and durable M1/M2 test evidence.
- Integrated RF Link Analysis with 3D Space View to visualize selected targets and orbital fleet nodes, orbital context, beam/link state, contact windows, link margin, payload effects, and mission closeout.

VisionGuard AI — Real-Time Warehouse Safety Intelligence

Problem solved: detection boxes alone cannot explain dangerous forklift-pedestrian trajectories or near-miss risks.

- Built a real-time vision pipeline with YOLO detection and ByteTrack tracking, then added virtual safety zones, homography-based real-world coordinates, distance, trajectory, speed, and closing-speed analysis.
- Implemented TTC/closest-approach prediction and explainable 0–100 risk scoring using distance, motion, shared zones, exposure duration, and detection confidence.
- Generated investigation-ready evidence for critical events, including pre/post-event video clips, snapshots, track IDs, risk reasons, timelines, and structured JSON metadata.

NeuroInsight AI — Agentic Clinical Intelligence for 3D MRI (Research Prototype)

Problem solved: medical-imaging predictions often lack anatomical evidence, measurements, supporting research, and review-ready reasoning.

- Designed a specialized-agent workflow spanning imaging inspection, segmentation, quantitative measurements, clinical findings, RAG evidence retrieval, case reasoning, structured reporting, and human review.
- Built a Flutter product experience for multi-planar and 3D MRI inspection, quantitative biomarkers, explainability heatmaps, case-aware assistance, retrieved evidence, and review-ready reports.
- Designed case-aware RAG to combine active-study findings with research and clinical knowledge while keeping retrieved evidence distinguishable from AI-generated synthesis.

EDUCATION

BS Software Engineering, *CGPA: 3.80* **| University of Okara** | 2020–2024

FSc Pre-Engineering | Punjab Group of Colleges

CERTIFICATIONS & ACHIEVEMENTS

University of Okara Nomination — Design & Development of Unmanned Aerial Systems
Samara National Research University, Russia | Nomination by the University of Okara | 2025

Prime Minister's Youth Laptop Scheme — Merit Recipient
Higher Education Commission (HEC), Government of Pakistan | 2023 | **Achievement Post**

Kaggle Notebook Expert Badge, Kaggle
Ranked Top 2,100 of ~60,000+ contributors in Kaggle Notebooks.

RESEARCH & PUBLICATIONS

3D-MobiBrainNet: Multi-class Alzheimer's disease classification using 3D brain magnetic resonance imaging — 27/08/2025
Ain Shams Engineering Journal, 16(11), 2025, Article 103714 | DOI: 10.1016/j.asej.2025.103714
Introduced a multi-plane 3D MRI classifier using depth-wise separable feature extraction, feature recalibration, and axial/coronal/sagittal fusion; achieved 97.33% accuracy and F1-score with 99.92% AUC on the ADNI dataset while emphasizing computational efficiency.

WIDENet: A Novel Lightweight CNN for Robust Skin Lesion Segmentation — 14/04/2026
IEEE Access, Volume 14, 2026, pp. 57763-57781 | DOI: 10.1109/ACCESS.2026.3683571
Developed a lightweight lesion-segmentation framework combining multi-resolution wavelet decomposition, transformer-based contextual encoding, and attentive feature refinement to preserve irregular lesion boundaries; evaluated across PH2, ISIC 2017, and ISIC 2018.

Hierarchical deep learning with stability-driven normalization and regularization for accurate multi-class diabetic retinopathy diagnosis — 21/09/2026
Ain Shams Engineering Journal, 17(12), 2026, Article 104453 | DOI: 10.1016/j.asej.2026.104453
Developed a compact hierarchical vision encoder for five-class diabetic-retinopathy grading, addressing class imbalance, subtle retinal features, and overfitting through stability-driven normalization and regularization; achieved 98.91% accuracy on APTOS with class-wise AUCs of 0.99-1.00.

Resilient UAV Network Security: Cross-Attentive Alignment of Sparse and Temporal Features for Adversarial Intrusion Detection — 28/10/2026
Vehicular Communications, 2026, Article 101094 | DOI: 10.1016/j.vehcom.2026.101094
Co-authored a TabNet–Transformer cross-attention framework with FGSM adversarial training; achieved 99.47%/98.24% accuracy on CIC-IDS2017/UNSW-NB15, +33.07% minority Botnet recall, and a 0.37–1.24 MB edge-ready footprint.